T4 Checkpoint 1: How each method calculates its result

Group T4, Tumor surface target prioritization (CPSC 4430). Code repository: https://github.com/JacksonMahr782/Machine-Learning

This sheet gives the equation behind every feature, label, model and metric in our plan. Part A covers the LUAD gene-dependency model. Part B covers the statistics in the TROP2 paper we reproduce.

Notation. g is a gene, s is a TCGA sample, ℓ is a DepMap cell line, n is the number of genes in a split, y_g is the true label and $\hat{y}_g$ is the model's prediction.

Part A. LUAD gene-dependency model (one row per gene)

A1. Features (from public data, never from DepMap)

Expression from Xena is $x_{gs} = \log_2(\mathrm{TPM}_{gs} + 1)$. T is the set of LUAD tumour samples and N is the set of normal lung samples.

Mean tumour expression (`expr_mean_tumor`):

$$\mu_g^T = \frac{1}{|T|} \sum_{s \in T} x_{gs}$$

Tumour-vs-normal enrichment (`expr_enrichment`):

$$e_g = \mu_g^T - \mu_g^N, \qquad \mu_g^N = \frac{1}{|N|} \sum_{s \in N} x_{gs}$$

Variability (`expr_sd_tumor`):

$$\sigma_g = \sqrt{\frac{1}{|T| - 1} \sum_{s \in T} (x_{gs} - \mu_g^T)^2}$$

The amplification features come from cBioPortal GISTIC calls, $c_{gs} \in \{-2, -1, 0, +1, +2\}$, over the S tumour samples. $\mathbf{1}[\cdot]$ is 1 when the condition is true and 0 otherwise.

Amplification frequency (`gistic_amp_freq`):

$$a_g = \frac{1}{|S|} \sum_{s \in S} \mathbf{1}[c_{gs} \geq 1]$$

High-level amplification frequency (`gistic_highamp_freq`):

$$h_g = \frac{1}{|S|} \sum_{s \in S} \mathbf{1}[c_{gs} = 2]$$

Surface flag (`is_surface`, from the Human Protein Atlas). It is used for eligibility and stratification only, never as a model input:

$$u_g = \mathbf{1}[\text{HPA main or additional location includes "Plasma membrane"}]$$

The feature vector for gene g is $\mathbf{x}_g = (\mu_g^T, e_g, \sigma_g, a_g, h_g)$.

A2. Label (from DepMap only)

L_g is the set of LUAD cell lines (Model.csv `OncotreeCode = LUAD`) with a non-missing Chronos effect $c_{g\ell}$ for gene g:

$$y_g = -\operatorname{median}_{\ell \in L_g} c_{g\ell} \quad \text{only if } |L_g| \geq 3; \text{ otherwise the gene is unlabeled.}$$

A higher y_g means LUAD cells depend on the gene more. Here 0 means no effect and about 1 is a typical essential gene. To prevent leakage, no other DepMap-derived value enters $\mathbf{x}_g$.

A3. Split and scaling

Labeled genes are split 70/15/15 into train, validation and test, stratified on u_g with seed 42. That gives every split the same share of surface genes. Each feature is standardized with the training mean and SD only:

$$z_{gj} = \frac{x_{gj} - \bar{x}_j^{\text{train}}}{s_j^{\text{train}}}$$

A4. Models

Constant baseline. It predicts the training mean for every gene:

$$\hat{y}_g = \bar{y}^{\text{train}} = \frac{1}{n_{\text{train}}} \sum_{g \in \text{train}} y_g$$

Ridge regression. It is a linear model with an L2 penalty. λ is chosen on the validation set:

$$\hat{y}_g = \beta_0 + \mathbf{z}_g^\top \boldsymbol{\beta}, \qquad \hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \sum_{g \in \text{train}} (y_g - \beta_0 - \mathbf{z}_g^\top \boldsymbol{\beta})^2 + \lambda \| \boldsymbol{\beta} \|_2^2$$

The penalized coefficients have a closed form, with y and Z centred:

$$\hat{\boldsymbol{\beta}} = (Z^\top Z + \lambda I)^{-1} Z^\top \mathbf{y}$$

Random forest. It averages B regression trees. Each tree T_b is grown on a bootstrap sample of training genes. At every split it picks the feature and threshold that most reduce the squared error inside the two child nodes:

$$\hat{y}_g = \frac{1}{B} \sum_{b=1}^{B} T_b(\mathbf{x}_g)$$

Gradient boosting. It adds small trees one at a time. Each new tree f_m is fitted to the current residuals, which are the negative gradient of the squared loss. v is the learning rate:

$$F_0 = \bar{y}^{\text{train}}, \qquad r_g^{(m)} = y_g - F_{m-1}(\mathbf{x}_g), \qquad F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + v f_m(\mathbf{x})$$

Training loss (all models). It is the mean squared error:

$$\text{MSE} = \frac{1}{n} \sum_g (y_g - \hat{y}_g)^2$$

A5. Evaluation metrics

Each metric is reported twice on the held-out test genes: once over all genes, and once over surface genes only ($u_g = 1$).

Mean absolute error (lower is better):

$$\mathrm{MAE} = \frac{1}{n}\sum_g |y_g - \hat{y}_g|$$

Spearman rank correlation (higher is better; it checks whether the model puts genes in the right order). With d_g the difference between gene g's rank by y and its rank by $\hat{y}$, and no ties:

$$\rho = 1 - \frac{6\sum_g d_g^2}{n(n^2 - 1)}$$

With ties, ρ is the Pearson correlation of the ranks.

A6. Improvement and ablation

Improvement. Expression × amplification interaction terms are added, so the model can learn that an amplified gene that is also highly expressed matters more:

$$\mathbf{x}_g^+ = (\mathbf{x}_g,\ \mu_g^T a_g,\ e_g a_g)$$

Amplification ablation. All amplification-derived columns (a_g, h_g and the interaction terms) are removed and the model is refitted on the same split. The effect of amplification is the change in test performance:

$$\Delta\mathrm{MAE} = \mathrm{MAE}_{\text{no amp}} - \mathrm{MAE}_{\text{full}}, \qquad \Delta\rho = \rho_{\text{full}} - \rho_{\text{no amp}}$$

Positive values mean amplification helps.

A7. Ranking surface candidates

Test genes with $u_g = 1$ (at least 20) are ranked with the course public-data guide's locked prioritization score. Predicted dependency $\hat{y}_g$ is one input and counts as supporting evidence only. Nonessential surface antigens stay eligible. The exact score formula is taken from the T4 project overview and is fixed before test genes are scored.

Part B. TROP2 paper reproduction (E-MTAB-16585, Figure 1)

B1. What we reproduce

This part reproduces two things. The first is TACSTD2 (TROP2) and LGR5 expression in each cell state. The second is the proportion of each cell state in each donor. The donor, not the cell, is the unit of analysis.

Per-cell normalized expression (typical single-cell pipeline; confirm the exact steps in Figure_01.ipynb). Here k_{ig} is the raw count of gene g in cell i:

$$\tilde{x}_{ig} = \log\left(1 + 10^4 \cdot \frac{k_{ig}}{\sum_{g'} k_{ig'}}\right)$$

Donor cell-state proportion. For donor d and state k:

$$p_{d,k} = \frac{n_{d,k}}{\sum_{k'} n_{d,k'}}$$

Here $n_{d,k}$ is the number of donor d's cells annotated as state k.

B2. Statistical tests the paper uses

Wilcoxon rank-sum test (comparing two groups). Rank all values together. R_1 is the sum of group 1's ranks:

$$U = R_1 - \frac{n_1 (n_1 + 1)}{2}$$

Benjamini–Hochberg correction (many genes tested at once). Sort the m p-values $p_{(1)} \le \ldots \le p_{(m)}$. Then reject every hypothesis up to the largest k with:

$$p_{(k)} \le \frac{k}{m} q$$

Log-rank test (survival curves in the 1,336-patient cohorts). At each event time j, O_{1j} is the observed number of deaths in group 1. The expected number is $E_{1j} = d_j \, n_{1j}/n_j$, and V_j is its variance:

$$\chi^2_{\mathrm{LR}} = \frac{(\sum_j (O_{1j} - E_{1j}))^2}{\sum_j V_j}, \qquad 1 \text{ degree of freedom}$$

Chi-squared test (counts in categories):

$$\chi^2 = \sum_{\text{cells}} \frac{(O - E)^2}{E}$$

Welch's t-test (two means):

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

Limiting-dilution assay (stem-cell frequency f). Under a single-hit Poisson model, the chance a well seeded with c cells shows no growth is:

$$P(\text{no growth}) = e^{-fc}$$

f is estimated by maximum likelihood. Groups are compared with a likelihood-ratio test.